

MCQS

BIO101 midterm preparation file

ZB

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In the name of God , Most
Gracious, Most merciful



BIO101

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MCQS

1. What is the basic unit of life?

- a) Atom
- b) Molecule
- c) Cell
- d) Tissue

Answer: c) Cell

2. Which of the following is a prokaryotic organism?

- a) Fungi
- b) Algae
- c) Bacteria
- d) Protozoa

Answer: c) Bacteria

3. Who is known as the father of modern taxonomy?

- a) Charles Darwin
- b) Gregor Mendel
- c) Carl Linnaeus
- d) Aristotle

Answer: c) Carl Linnaeus

4. Which macromolecule is primarily responsible for storing genetic information?

- a) Proteins
- b) Lipids
- c) DNA
- d) Carbohydrates

Answer: c) DNA

5. What type of bond forms between water molecules?

- a) Ionic bond
- b) Hydrogen bond
- c) Covalent bond
- d) Metallic bond

Answer: b) Hydrogen bond

6. Which organelle is responsible for energy production in eukaryotic cells?

- a) Nucleus
- b) Ribosomes
- c) Mitochondria
- d) Golgi apparatus

Answer: c) Mitochondria

7. What is the function of ribosomes in a cell?

- a) Protein synthesis
- b) DNA replication
- c) Energy production
- d) Transport of molecules

Answer: a) Protein synthesis

8. What is the main function of the cell membrane?

- a) To provide energy
- b) To control the movement of substances in and out of the cell
- c) To synthesize proteins
- d) To store genetic information

Answer: b) To control the movement of substances in and out of the cell

9. Which of the following structures is found in plant cells but not in animal cells?

- a) Mitochondria
- b) Nucleus
- c) Chloroplast
- d) Ribosomes

Answer: c) Chloroplast

10. What is the process by which cells divide to form two identical daughter cells?

- a) Mitosis
- b) Meiosis
- c) Binary fission
- d) Budding

Answer: a) Mitosis

11. Which of the following is the energy currency of the cell?

- a) NADPH
- b) ATP
- c) Glucose
- d) RNA

Answer: b) ATP

12. What is the primary function of the endoplasmic reticulum?

- a) Digestion
- b) Protein and lipid synthesis
- c) Genetic storage
- d) Photosynthesis

Answer: b) Protein and lipid synthesis

13. Which of the following is a characteristic of all living organisms?

- a) Ability to reproduce
- b) Ability to photosynthesize
- c) Ability to fly
- d) Ability to swim

Answer: a) Ability to reproduce

14. Which organelle is responsible for packaging and modifying proteins?

- a) Nucleus
- b) Ribosomes
- c) Golgi apparatus
- d) Lysosomes

Answer: c) Golgi apparatus

15. What is the function of lysosomes in a cell?

- a) Protein synthesis
- b) Detoxification
- c) Digestion of waste materials
- d) Lipid synthesis

Answer: c) Digestion of waste materials

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ZB

16. What is the structure that carries genetic information in eukaryotic cells?

- a) Chromosomes
- b) Mitochondria
- c) Cytoplasm
- d) Ribosomes

Answer: a) Chromosomes

17. Which molecule carries amino acids to the ribosome for protein synthesis?

- a) mRNA
- b) tRNA
- c) rRNA
- d) DNA

Answer: b) tRNA

18. What is the name of the process by which plants make their own food using sunlight?

- a) Photosynthesis
- b) Cellular respiration
- c) Fermentation
- d) Digestion

Answer: a) Photosynthesis

19. Which of the following is not part of the cell theory?

- a) All living things are made of cells
- b) The cell is the basic unit of life
- c) All cells come from pre-existing cells
- d) All cells have a nucleus

Answer: d) All cells have a nucleus

20. What is the function of the mitochondria?

- a) Energy production
- b) Protein synthesis
- c) Genetic storage
- d) Detoxification

Answer: a) Energy production

21. In which phase of the cell cycle does DNA replication occur?

- a) G1 phase**
- b) S phase**
- c) G2 phase**
- d) M phase**

Answer: b) S phase

22. Which of the following is a feature of prokaryotic cells?

- a) Nucleus**
- b) Membrane-bound organelles**
- c) Lack of a nucleus**
- d) Presence of mitochondria**

Answer: c) Lack of a nucleus

23. Which process is responsible for the movement of water across a cell membrane?

- a) Diffusion**
- b) Active transport**
- c) Osmosis**
- d) Endocytosis**

Answer: c) Osmosis

24. What type of cell division results in four non-identical daughter cells?

- a) Mitosis**
- b) Meiosis**
- c) Binary fission**
- d) Budding**

Answer: b) Meiosis

25. What is the function of the nucleolus?

- a) Protein synthesis**
- b) Storage of RNA**
- c) Production of ribosomes**
- d) Packaging of proteins**

Answer: c) Production of ribosomes

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26. Which structure is found in both plant and animal cells?

- a) Chloroplast
- b) Nucleus
- c) Cell wall
- d) Vacuole

Answer: b) Nucleus

27. What is the function of chlorophyll in plant cells?

- a) Absorbs water
- b) Absorbs light energy for photosynthesis
- c) Synthesizes glucose
- d) Converts glucose into ATP

Answer: b) Absorbs light energy for photosynthesis

28. Which macromolecule is made up of amino acids?

- a) Carbohydrates
- b) Nucleic acids
- c) Proteins
- d) Lipids

Answer: c) Proteins

29. Which of the following is a type of passive transport?

- a) Active transport
- b) Diffusion
- c) Endocytosis
- d) Phagocytosis

Answer: b) Diffusion

30. What is the function of the Golgi apparatus?

- a) Energy production
- b) Protein modification and packaging
- c) DNA replication
- d) Lipid synthesis

Answer: b) Protein modification and packaging

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31. Which of the following is an example of a lipid?

- a) Glucose
- b) Protein
- c) Fatty acids
- d) Nucleotides

Answer: c) Fatty acids

32. What is the name of the structure that helps cells maintain shape and internal organization?

- a) Cytoskeleton
- b) Nucleus
- c) Ribosomes
- d) Mitochondria

Answer: a) Cytoskeleton

33. What is the primary function of white blood cells?

- a) Oxygen transport
- b) Defense against infection
- c) Clotting
- d) Hormone production

Answer: b) Defense against infection

34. What is the function of the vacuole in plant cells?

- a) Storage of water and nutrients
- b) Protein synthesis
- c) Energy production
- d) Cell division

Answer: a) Storage of water and nutrients

35. Which of the following is a function of the smooth endoplasmic reticulum?

- a) Protein synthesis
- b) Lipid synthesis
- c) ATP production
- d) Photosynthesis

Answer: b) Lipid synthesis

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36. Which of the following structures is responsible for cell movement?

- a) Nucleus
- b) Cytoskeleton
- c) Ribosomes
- d) Mitochondria

Answer: b) Cytoskeleton

37. What is the function of centrioles?

- a) Help in cell division
- b) Protein synthesis
- c) Detoxification
- d) Storage of genetic material

Answer: a) Help in cell division

38. What does a gene consist of?

- a) Amino acids
- b) RNA
- c) DNA sequence
- d) Ribosomes

Answer: c) DNA sequence

39. Which of the following is an example of a carbohydrate?

- a) Glucose
- b) Fatty acid
- c) Protein
- d) Nucleotide

Answer: a) Glucose

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41. What is the function of DNA in a cell?

- a) Store genetic information**
- b) Synthesize proteins**
- c) Store energy**
- d) Provide structure to the cell**

Answer: a) Store genetic information

42. Which of the following is not a function of proteins?

- a) Store energy**
- b) Catalyze reactions**
- c) Provide structure**
- d) Transport molecules**

Answer: a) Store energy

43. What is the function of the cell wall in plant cells?

- a) Provide structure and protection**
- b) Transport nutrients**
- c) Synthesize proteins**
- d) Store water**

Answer: a) Provide structure and protection

44. Which of the following is a function of the plasma membrane?

- a) Transmit signals to the nucleus**
- b) Maintain the cell's shape**
- c) Control the entry and exit of materials**
- d) Synthesize proteins**

Answer: c) Control the entry and exit of materials

45. Which of the following is a characteristic of eukaryotic cells?

- a) Lack of nucleus**
- b) Presence of membrane-bound organelles**
- c) No ribosomes**
- d) No cell membrane**

Answer: b) Presence of membrane-bound organelles

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51. In photosynthesis, what molecule is split to release oxygen?

- a) Water
- b) Carbon dioxide
- c) Glucose
- d) Oxygen

Answer: a) Water

52. What is the primary product of the light-dependent reactions in photosynthesis?

- a) Glucose
- b) Oxygen
- c) ATP and NADPH
- d) Carbon dioxide

Answer: c) ATP and NADPH

53. What process converts glucose into pyruvate?

- a) Citric acid cycle
- b) Glycolysis
- c) Fermentation
- d) Electron transport chain

Answer: b) Glycolysis

54. Which molecule is the main source of energy for cellular activities?

- a) Glucose
- b) ATP
- c) NADPH
- d) Fats

Answer: b) ATP

55. In the process of DNA replication, which enzyme is responsible for unwinding the DNA double helix?

- a) Ligase
- b) DNA polymerase
- c) Helicase
- d) RNA polymerase

Answer: c) Helicase

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56. Which of the following is the site of protein synthesis?

- a) Nucleus
- b) Golgi apparatus
- c) Ribosome
- d) Mitochondria

Answer: c) Ribosome

57. During which phase of mitosis do sister chromatids separate?

- a) Prophase
- b) Anaphase
- c) Metaphase
- d) Telophase

Answer: b) Anaphase

58. What is the primary function of the smooth endoplasmic reticulum?

- a) Protein synthesis
- b) Lipid synthesis
- c) Detoxification
- d) Both b and c

Answer: d) Both b and c

59. Which of the following is a type of RNA involved in protein synthesis?

- a) mRNA
- b) rRNA
- c) tRNA
- d) All of the above

Answer: d) All of the above

60. Which of the following are the building blocks of proteins?

- a) Nucleotides
- b) Fatty acids
- c) Amino acids
- d) Monosaccharides

Answer: c) Amino acids

61. What is the function of the Golgi apparatus?

- a) Synthesize proteins**
- b) Store energy**
- c) Modify, sort, and package proteins**
- d) Control cell activities**

Answer: c) Modify, sort, and package proteins

62. In which organelle does photosynthesis occur?

- a) Mitochondrion**
- b) Chloroplast**
- c) Nucleus**
- d) Ribosome**

Answer: b) Chloroplast

63. What is the main component of the plasma membrane?

- a) Proteins**
- b) Nucleic acids**
- c) Carbohydrates**
- d) Phospholipids**

Answer: d) Phospholipids

64. Which of the following is a function of the large central vacuole in plant cells?

- a) Store water**
- b) Store food**
- c) Store waste products**
- d) All of the above**

Answer: d) All of the above

65. Which of the following molecules is produced by photosynthesis and used in cellular respiration?

- a) Oxygen**
- b) Carbon dioxide**
- c) Glucose**
- d) ATP**

Answer: c) Glucose

66. Which of the following is a characteristic of asexual reproduction?

- a) Involves two parents**
- b) Results in offspring that are genetically identical to the parent**
- c) Involves meiosis**
- d) Produces offspring with unique genetic combinations**

Answer: b) Results in offspring that are genetically identical to the parent

67. In the process of meiosis, how many daughter cells are produced?

- a) Two**
- b) Four**
- c) Six**
- d) Eight**

Answer: b) Four

68. What is the function of the spindle fibers during cell division?

- a) To attach to the cell membrane**
- b) To help in the formation of the nuclear envelope**
- c) To separate the chromosomes**
- d) To package the DNA**

Answer: c) To separate the chromosomes

69. Which of the following is true about the electron transport chain in cellular respiration?

- a) It occurs in the cytoplasm**
- b) It produces oxygen**
- c) It occurs in the mitochondria**
- d) It directly produces ATP**

Answer: c) It occurs in the mitochondria

70. What is the process by which a cell engulfs large particles?

- a) Exocytosis**
- b) Endocytosis**
- c) Diffusion**
- d) Osmosis**

Answer: b) Endocytosis

71. What are the structures that help prokaryotic cells attach to surfaces?

- a) Pili**
- b) Flagella**
- c) Cilia**
- d) Ribosomes**

Answer: a) Pili

72. What is the role of telomerase in eukaryotic cells?

- a) To replicate the DNA**
- b) To shorten telomeres**
- c) To prevent DNA damage**
- d) To maintain the length of telomeres**

Answer: d) To maintain the length of telomeres

73. Which of the following is true about enzymes?

- a) They are consumed in a chemical reaction**
- b) They lower the activation energy of a reaction**
- c) They increase the activation energy of a reaction**
- d) They are carbohydrates**

Answer: b) They lower the activation energy of a reaction

74. What is the name of the process in which mRNA is converted into a protein?

- a) Transcription**
- b) Replication**
- c) Translation**
- d) Elongation**

Answer: c) Translation

75. Which of the following describes the function of the cytoplasm?

- a) Store DNA**
- b) Facilitate protein synthesis**
- c) Store water**
- d) Provide structure and support to the cell**

Answer: d) Provide structure and support to the cell

76. Which structure is responsible for maintaining cell shape and assisting with cell movement?

- a) Nucleus**
- b) Cytoskeleton**
- c) Golgi apparatus**
- d) Ribosomes**

Answer: b) Cytoskeleton

77. Which of the following is a characteristic of meiosis?

- a) Produces genetically identical cells**
- b) Occurs in somatic cells**
- c) Involves two rounds of division**
- d) Results in two daughter cells**

Answer: c) Involves two rounds of division

78. What is the primary role of the small subunit of a ribosome?

- a) Synthesizes RNA**
- b) Attaches to the mRNA**
- c) Catalyzes the formation of peptide bonds**
- d) Stores genetic information**

Answer: b) Attaches to the mRNA

79. Which molecule is responsible for carrying the genetic code from DNA to the ribosome?

- a) tRNA**
- b) mRNA**
- c) rRNA**
- d) DNA**

Answer: b) mRNA

80. Which of the following statements about the cell cycle is correct?

- a) The cell cycle consists of interphase and mitosis**
- b) Mitosis is the longest phase of the cell cycle**
- c) The cell cycle occurs only in eukaryotic cells**
- d) The cell cycle does not involve the replication of DNA**

Answer: a) The cell cycle consists of interphase and mitosis

MIDTERM Important topic from 1 to 22 lectures

Phylum Echinodermata

Echinoderms include common seashore animals such as sea stars (also known as "starfish"), sand dollars, and sea urchins, along with hundreds of more exotic forms. An echinoderm is any member of the phylum Echinodermata of marine animals. The adults are recognizable by their radial symmetry, and include starfish, sea urchins, sand dollars, sea cucumbers, and sea lilies or "stone lilies."

Phylum Chordata

The Phylum Chordata includes the well-known vertebrates (fishes, amphibians, reptiles, birds, mammals). A chordate is an animal of the phylum Chordata. All chordates possess 5 synapomorphies, or primary characteristics, at some point during their larval or adulthood stages that distinguish them from all other animals.

Three main characteristics of chordates:

1. Chordata is the most advanced animal phylum.
2. A notochord, a solid unjointed rod, is present in chordates at some stages of life.
3. A dorsal hollow nerve cord develops in the embryo and usually persists throughout life in most chordates.
4. Gill slits are present at some stages of life.

Characteristics of Urochordata

Possesses a notochord, a hollow nerve cord, and a post-anal tail.

Body has more than two cell layers and includes tissues and organs.

Has a U-shaped gut.

Body has no coelomic body cavity.

Body wholly enclosed in a 'tunic' of secreted protein and cellulose-like material.

What is chordate metamerism?

In chordates, metamerism evolved independently for locomotion. Metamerism allowed the tail muscles to be arranged segmentally for the undulatory movement of the body.

Animals belonging to the subphylum Cephalochordates

Cephalochordates include animals that possess all 5 synapomorphies, such as the notochord, dorsal hollow nerve cord, endostyle, pharyngeal slits, and a post-anal tail. An example of Cephalochordates is the Bahamas lancelet, *Asymmetron lucayanum*.

Sub-phylum Vertebrata General Characteristics

Diversity of vertebrates: The subphylum Vertebrata contains all animals that possess backbones, gills, and a central nervous system in at least one phase of development. Vertebrates include amphibians, reptiles, mammals, and birds, as well as jawless fishes, bony fishes, sharks, and rays.

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What does Gymnosperm mean?

A gymnosperm is any member of the class or subdivision Gymnospermae, a group of woody vascular seed plants (such as conifers or cycads) that produce naked seeds not enclosed in an ovary. Some gymnosperms have motile spermatozoids.

Angiosperms Definition

An angiosperm is a plant that produces flowers. Angiosperms, also known as flowering plants, are a vital group of plants having seeds.

Definition of monocot

A monocot is a chiefly herbaceous angiospermous plant (such as a grass, lily, or palm) with a single cotyledon in the embryo, parallel-veined leaves, and floral organs arranged in multiples of three. Monocots account for a quarter of all flowering plants.

Dicots

A dicot, or dicotyledon, is a member of the flowering plants (angiosperms) that has a pair of leaves, or cotyledons, in the embryo of the seed.

Sexual Reproduction in Plants

Flowering plants reproduce sexually through pollination. Flowers contain male sex organs called stamens and female sex organs called pistils. The anther is the part of the stamen that contains pollen. Self-pollination happens when a plant's own pollen fertilizes its own ovules.

Development of the Ovule and Female Gamete

The female gametophyte develops within the ovule and generally consists of three antipodal cells, one central cell, two synergid cells, and one egg cell. The female gametophyte is also called the embryo sac or megagametophyte.

Evolution of Gymnosperms

Gymnosperms were the first seed plants to evolve. The earliest seed-like bodies are found in rocks of the Upper Devonian Series (about 382.7 million to 358.9 million years ago).

Life Cycle of Gymnosperms

The life cycle of a gymnosperm involves alternation of generations, with a dominant sporophyte in which reduced male and female gametophytes reside. All gymnosperms are heterosporous, with male and female spores developing in different strobili.

Cycads

Cycads (order Cycadales) are an ancient group of palmlike gymnosperms with about 300 extant species across three families. They are distinguished by crowns of large pinnately compound leaves and by large cones.

Finall term preparation

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



**"Marks ka tension? Bas
apne dil ko samjhao: 'Main
genius hoon!'"**

Final term preparation

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1. Which of the following is the basic unit of life?

- a) Organ
- b) Cell
- c) Tissue
- d) Organism

Answer: b) Cell

2. Which part of the cell is responsible for energy production?

- a) Nucleus
- b) Mitochondria
- c) Ribosomes
- d) Endoplasmic reticulum

Answer: b) Mitochondria

3. Which of the following is not a type of RNA?

- a) mRNA
- b) tRNA
- c) rRNA
- d) dRNA

Answer: d) dRNA

4. Which process occurs in the nucleus of a eukaryotic cell?

- a) Transcription
- b) Translation
- c) Glycolysis
- d) Krebs cycle

Answer: a) Transcription

5. Which of the following is true about enzymes?

- a) Enzymes are consumed during reactions.
- b) Enzymes speed up chemical reactions.
- c) Enzymes increase the activation energy.
- d) Enzymes are only proteins.

Answer: b) Enzymes speed up chemical reactions.

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6. Which of the following is the primary source of energy for living organisms?

- a) Lipids
- b) Proteins
- c) Carbohydrates
- d) Nucleic acids

Answer: c) Carbohydrates

7. What is the process by which cells divide to form two identical daughter cells?

- a) Meiosis
- b) Binary fission
- c) Mitosis
- d) Cytokinesis

Answer: c) Mitosis

8. Which molecule carries genetic information?

- a) Protein
- b) RNA
- c) DNA
- d) Lipid

Answer: c) DNA

9. The structure that separates the inside of the cell from the external environment is called the:

- a) Ribosome
- b) Plasma membrane
- c) Nucleus
- d) Mitochondrion

Answer: b) Plasma membrane

10. Which of the following best describes a codon?

- a) A sequence of amino acids
- b) A sequence of three nucleotides on mRNA
- c) A sequence of proteins
- d) A sequence of tRNA molecules

Answer: b) A sequence of three nucleotides on mRNA

11. Which of the following is not part of the cell theory?

- a) All living organisms are made of cells.
- b) Cells are the basic unit of life.
- c) All cells arise from pre-existing cells.
- d) Cells have a nucleus.

Answer: d) Cells have a nucleus.

12. Which organelle is responsible for protein synthesis?

- a) Mitochondrion
- b) Nucleus
- c) Ribosome
- d) Golgi apparatus

Answer: c) Ribosome

13. Which of the following processes does not require oxygen?

- a) Aerobic respiration
- b) Photosynthesis
- c) Anaerobic respiration
- d) Electron transport chain

Answer: c) Anaerobic respiration

14. Which of the following molecules is a monosaccharide?

- a) Glucose
- b) Sucrose
- c) Lactose
- d) Starch

Answer: a) Glucose

15. What is the main function of the lysosome?

- a) Protein synthesis
- b) Cellular respiration
- c) Digestion of cellular waste
- d) Storage of genetic material

Answer: c) Digestion of cellular waste

16. Which of the following is responsible for transporting proteins within the cell?

- a) Endoplasmic reticulum**
- b) Mitochondria**
- c) Ribosomes**
- d) Golgi apparatus**

Answer: a) Endoplasmic reticulum

17. Which of the following occurs during prophase in mitosis?

- a) Chromosomes align at the cell's equator**
- b) Nuclear membrane disappears**
- c) Sister chromatids separate**
- d) DNA replicates**

Answer: b) Nuclear membrane disappears

18. Which of the following is a characteristic of prokaryotic cells?

- a) Membrane-bound organelles**
- b) A nucleus**
- c) A single circular chromosome**
- d) Multiple linear chromosomes**

Answer: c) A single circular chromosome

19. What is the role of tRNA in protein synthesis?

- a) Carries the genetic code**
- b) Assembles amino acids into proteins**
- c) Transcribes DNA into mRNA**
- d) Brings amino acids to the ribosome**

Answer: d) Brings amino acids to the ribosome

20. Which of the following is a function of the Golgi apparatus?

- a) Protein synthesis**
- b) Sorting and packaging proteins**
- c) Cellular respiration**
- d) Photosynthesis**

Answer: b) Sorting and packaging proteins

Mcqs**Final term preparation**

21. Which of the following is a function of the mitochondria?

- a) DNA replication**
- b) Cellular respiration and energy production**
- c) Protein synthesis**
- d) Storage of lipids**

Answer: b) Cellular respiration and energy production

22. Which of the following is true about homologous chromosomes?

- a) They are identical copies of a chromosome.**
- b) They carry the same genes, but possibly different alleles.**
- c) They are found only in gametes.**
- d) They do not pair during meiosis.**

Answer: b) They carry the same genes, but possibly different alleles.

23. Which type of bond holds the two strands of DNA together?

- a) Hydrogen bonds**
- b) Ionic bonds**
- c) Covalent bonds**
- d) Peptide bonds**

Answer: a) Hydrogen bonds

24. In which part of the cell does the citric acid cycle occur?

- a) Cytoplasm**
- b) Mitochondrial matrix**
- c) Nucleus**
- d) Endoplasmic reticulum**

Answer: b) Mitochondrial matrix

25. Which of the following is a result of meiosis?

- a) Two diploid cells**
- b) Four haploid cells**
- c) Two haploid cells**
- d) Four diploid cells**

Answer: b) Four haploid cells

Mcqs

Final term preparation

ZB

26. Which of the following is an example of passive transport?

- a) Endocytosis
- b) Facilitated diffusion
- c) Active transport
- d) Exocytosis

Answer: b) Facilitated diffusion

27. What is the final product of glycolysis?

- a) Oxygen
- b) Glucose
- c) Pyruvate
- d) NADH

Answer: c) Pyruvate

28. Which process converts light energy into chemical energy?

- a) Respiration
- b) Photosynthesis
- c) Fermentation
- d) Glycolysis

Answer: b) Photosynthesis

29. Which of the following is the main component of the cell membrane?

- a) Proteins
- b) Carbohydrates
- c) Phospholipids
- d) Nucleic acids

Answer: c) Phospholipids

30. Which of the following is true about the function of ribosomes?

- a) Ribosomes are responsible for DNA replication.
- b) Ribosomes synthesize proteins.
- c) Ribosomes are involved in cellular respiration.
- d) Ribosomes break down proteins.

Answer: b) Ribosomes synthesize proteins

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31. Which of the following best describes the function of the mitochondrion?

- a) Protein synthesis
- b) Energy production
- c) Lipid storage
- d) Waste breakdown

Answer: b) Energy production

32. Which of the following organelles contains digestive enzymes?

- a) Nucleus
- b) Lysosome
- c) Golgi apparatus
- d) Endoplasmic reticulum

Answer: b) Lysosome

33. Which of the following is a product of the Calvin cycle in photosynthesis?

- a) Oxygen
- b) Glucose
- c) ATP
- d) NADH

Answer: b) Glucose

34. Which of the following is true about the relationship between DNA and RNA?

- a) RNA is synthesized from DNA.
- b) RNA and DNA have the same base pairs.
- c) RNA is more stable than DNA.
- d) DNA is synthesized from RNA.

Answer: a) RNA is synthesized from DNA.

35. Which of the following is NOT a part of the process of protein synthesis?

- a) Transcription
- b) Translation
- c) Replication
- d) Splicing

Answer: c) Replication

Mcqs

Final term preparation

36. What is the primary purpose of the light-dependent reactions in photosynthesis?

- a) To produce glucose
- b) To produce oxygen
- c) To capture light energy and convert it into chemical energy
- d) To fix carbon into organic compounds

Answer: c) To capture light energy and convert it into chemical energy

37. Which of the following statements is true about prokaryotic cells?

- a) They have a membrane-bound nucleus.
- b) They lack a plasma membrane.
- c) They contain ribosomes but no membrane-bound organelles.
- d) They are always multicellular.

Answer: c) They contain ribosomes but no membrane-bound organelles.

38. Which of the following is a key characteristic of enzymes?

- a) They can function at a wide range of temperatures.
- b) They are consumed in chemical reactions.
- c) They lower the activation energy of reactions.
- d) They are usually carbohydrates.

Answer: c) They lower the activation energy of reactions.

39. What is the final electron acceptor in the electron transport chain?

- a) NADH
- b) Oxygen
- c) Glucose
- d) ATP

Answer: b) Oxygen

40. Which molecule is directly responsible for carrying amino acids to the ribosome during translation?

- a) mRNA
- b) rRNA
- c) tRNA
- d) DNA

Answer: c) tRNA

Mcqs

Final term preparation

ZB

41. What is the main function of the smooth endoplasmic reticulum?

- a) Protein synthesis
- b) Lipid synthesis and detoxification
- c) Packaging proteins
- d) Storing genetic information

Answer: b) Lipid synthesis and detoxification

42. Which of the following is an example of active transport?

- a) Osmosis
- b) Facilitated diffusion
- c) Sodium-potassium pump
- d) Simple diffusion

Answer: c) Sodium-potassium pump

43. Which phase of the cell cycle is responsible for cell growth and DNA replication?

- a) G1 phase
- b) S phase
- c) G2 phase
- d) M phase

Answer: b) S phase

44. Which of the following occurs during telophase of mitosis?

- a) Chromosomes condense
- b) The nuclear envelope reforms
- c) Chromatids separate
- d) DNA replication occurs

Answer: b) The nuclear envelope reforms

45. Which of the following best describes a gene?

- a) A sequence of amino acids
- b) A segment of DNA that codes for a protein
- c) A structure in the cytoplasm
- d) A single nucleotide

Answer: b) A segment of DNA that codes for a protein

46. Which of the following is NOT a component of a nucleotide?

- a) Phosphate group**
- b) Ribose sugar**
- c) Nitrogenous base**
- d) Amino acid**

Answer: d) Amino acid

47. In which part of the cell does translation occur?

- a) Nucleus**
- b) Ribosome**
- c) Mitochondrion**
- d) Endoplasmic reticulum**

Answer: b) Ribosome

48. What is the main function of the cell wall in plant cells?

- a) Protein synthesis**
- b) Regulating cell division**
- c) Protecting and supporting the cell**
- d) Storing genetic information**

Answer: c) Protecting and supporting the cell

49. What is the difference between a homozygous and a heterozygous individual?

- a) Homozygous individuals have two different alleles.**
- b) Heterozygous individuals have two identical alleles.**
- c) Homozygous individuals have two identical alleles.**
- d) There is no difference between them.**

Answer: c) Homozygous individuals have two identical alleles.

50. Which of the following is the main function of the Golgi apparatus?

- a) Synthesis of lipids**
- b) Protein modification and packaging**
- c) Breakdown of cellular waste**
- d) Energy production**

Answer: b) Protein modification and packaging

51. Which of the following is NOT a type of RNA?

- a) mRNA**
- b) rRNA**
- c) tRNA**
- d) qRNA**

Answer: d) qRNA

52. In which part of the chloroplast does the Calvin cycle occur?

- a) Stroma**
- b) Thylakoid membrane**
- c) Cytoplasm**
- d) Mitochondrion**

Answer: a) Stroma

53. What type of bond holds together the two strands of DNA?

- a) Ionic bonds**
- b) Hydrogen bonds**
- c) Covalent bonds**
- d) Peptide bonds**

Answer: b) Hydrogen bonds

54. Which of the following is true about RNA polymerase?

- a) It synthesizes proteins.**
- b) It synthesizes RNA from a DNA template.**
- c) It is involved in the Calvin cycle.**
- d) It binds to ribosomes.**

Answer: b) It synthesizes RNA from a DNA template.

55. Which of the following is true about the structure of DNA?

- a) It is a single strand of nucleotides.**
- b) It has a double helix shape.**
- c) It is composed of ribose sugars.**
- d) It contains uracil instead of thymine.**

Answer: b) It has a double helix shape

56. What is the primary role of the centrioles during mitosis?

- a) To form the mitotic spindle**
- b) To synthesize proteins**
- c) To replicate DNA**
- d) To break down the nuclear envelope**

Answer: a) To form the mitotic spindle

57. What is the end product of the Krebs cycle?

- a) Glucose**
- b) ATP**
- c) Oxygen**
- d) Carbon dioxide**

Answer: d) Carbon dioxide

58. Which of the following is NOT a phase of mitosis?

- a) Prophase**
- b) Metaphase**
- c) Telophase**
- d) Interphase**

Answer: d) Interphase

59. Which of the following is true regarding sexual reproduction?

- a) It produces genetically identical offspring.**
- b) It requires two parents.**
- c) It produces offspring that are clones of the parent.**
- d) It occurs without the involvement of gametes.**

Answer: b) It requires two parents.

60. Which of the following is a characteristic of asexual reproduction?

- a) Involves two parents**
- b) Results in genetically diverse offspring**
- c) Requires gametes**
- d) Results in genetically identical offspring**

Answer: d) Results in genetically identical offspring

MCQS

Midterm preparation file

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.

Therefore, for passing a course/subject, student shall fulfill the following:

- Secure minimum **20%** score in Formative Assessments/Mid terms
- Secure minimum **20%** score in Final Term Examinations.
- Secure at least **40%** marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?

A: Follow these steps:

- 1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).**
- 2. Each subject typically has 3 credit hours.**
- 3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.**

Examples:

21 credit hours = 7 subjects

15 credit hours = 5 subjects

Remember, the university may allow different credit hours for each student, so check your specific allowance.

By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed

Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish
krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat
Fame

Moat

**Humy pta hona chaheay Moat tab ani ha jab Allah
chahy us se pehly puri dunya bi ak taraf ho ke apko
marna chahy to apka kuch nahe beggar saqti**

Risk/Dolat

**Risk Dolat sb Allah pak ke hath m ha puri dunya ak
taraf b ho jay na apsy apka risk cheen saqti ha na de
saqti ha Ye sirf ALLAH pak ky hath m ha**

Izat Zilat shohrat Fame

**Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b
apko zra brabr b damage nahe kr saqti Agr ALLAH**

**Apko izat dena chahay to puri dunya ak taraf ho ke
b apko 1% b nuksan nahe pohancha saqti**

Phr Dar ks bat ka ????????????????

**Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne
mat jhuko puri dunya mel k b apka kuch b nahe
beggar saqti**

Tips

Payary Students kbi b Only files pe depend mat kea kryn just files read krke uspe total depend krna it's not Good Vu ke tamat Papers Conceptual hoty hn Agr apky concepts

Apny concepts clear kryn or handout ko read kryn

Agr Handouts lengthy

Lgy to Ap har topics Ki basic definitions dekh lea kryn or headings dekh leay kryn

Ap thori se mehnat Kryn bs ap kbi b fail nahe hongy jab himat haar jao to un maa baap k bare m socha kro Jo apki fees dete hn jinko ap pe bohat dawy hn Jo har jga kehty hn Hamara bacha university Mai parh rha ha un maa baap ki umeedy Kahi mat Tory apny kea naa Sahi apny maa baap k leay mehnat kro himat kro Everything is possible just do it my Family

Apko Kuch b samj naa ay yaa ap Kuch b pochna chahy to ap mujsy contact kr saqty hn

> ZB

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Mri tamaat files Assignment GDB Solutions files Quiz etc. A to Z Sb free of cost ha Vu k bohat se log mere name pe passy lete hn Mera unsy koi lena dena nahe so be aware from them ZB Jo b krta ha Allah ki mehlok ki khudtam k leay krta ha I don't need anything from anyone in return because I am doing this business with My Allah

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family

Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my campaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha

Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us

m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us

Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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ZB

Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

..... **MY Family**

Dear students, all Virtual University groups, from A to Z, have been created by students themselves. Virtual University does not have any official groups. If anyone claims their group is official, they are fake. Please be cautious. Virtual University has not yet made any announcement regarding an official WhatsApp group.

..... **AL-Quran**

"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"

May Allah bless you